Upper and lower bounds

The national revision and practical lessons, carrying the Cambridge Stage 9 treatment of bounds in calculations.

LESSON 58–59 | 2 × 40 MIN | ALGEBRA 8, CHAPTER II REVISION | STAGE 9 · 3.4 [CAMBRIDGE INSERT]

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Write the upper and lower bound of a rounded measurement.
- Find the bounds of a sum and of a difference.
- Find the bounds of a product and of a quotient.
- Decide how many figures an answer may honestly be given to.

TEXTBOOK

National	pp. 118–121
Cambridge	Learner's Book pp. 76–81
Workbook	Workbook 3.4

01 Explanation

The bounds of a rounded number

RULE

A value rounded to the nearest u lies within $\frac{u}{2}$ of the stated figure:

$$x - \frac{u}{2} \leq \textit{true value} < x + \frac{u}{2}$$

STATED	ROUNDED TO	LOWER BOUND	UPPER BOUND
7 cm	nearest cm	6.5	7.5
4.8 m	1 d.p.	4.75	4.85
250 g	nearest 10 g	245	255
3.60 s	2 d.p.	3.595	3.605

The lower bound is reached; the upper bound is not. That is why it is written with $\leq$ on the left and $<$ on the right.

Bounds of a sum, difference, product, quotient

WHICH BOUND WITH WHICH

	LARGEST ANSWER	SMALLEST ANSWER
$a + b$	$UB(a) + UB(b)$	$LB(a) + LB(b)$
$a - b$	$UB(a) - LB(b)$	$LB(a) - UB(b)$
$a \times b$	$UB(a) \times UB(b)$	$LB(a) \times LB(b)$
$a \div b$	$UB(a) \div LB(b)$	$LB(a) \div UB(b)$

(for positive quantities)

The pattern is worth saying aloud: for a **sum** or a **product**, both go the same way; for a **difference** or a **quotient**, they go opposite ways — to make a difference as large as possible, take the most you can and subtract the least you can.

How many figures may the answer have?

Work out both bounds, then keep only the digits on which they agree.

If the bounds of an area come out as 14.3175 and 15.0875, the two do not even agree on the units digit — so the honest answer is “about 15”, not $14.7\ m^2$. Reporting more figures than the bounds support is the commonest fault in practical work.

02 Worked examples

EXAMPLE 1 **A rectangle measures 4.2 m by 3.5 m, each to 1 d.p. Find the bounds of the perimeter.**

1. $4.15 \le a < 4.25$ and $3.45 \le b < 3.55$

   Half of 0.1 either way.

2. $P = 2(a + b)$

3. $LB = 2(4.15 + 3.45) = 15.2$

   Both lower bounds.

4. $UB = 2(4.25 + 3.55) = 15.6$

   Both upper bounds.

ANSWER

$15.2\ m \le P < 15.6\ m$

EXAMPLE 2 **The same rectangle: find the bounds of the area.**

1. $LB = 4.15 \times 3.45 = 14.3175$

   Both lower bounds.

2. $UB = 4.25 \times 3.55 = 15.0875$

   Both upper bounds.

3. The bounds disagree already in the units digit.

4. So the area is “about 15 m²” — one significant figure is all that is justified.

ANSWER

$14.32\ m^2 \le S < 15.09\ m^2$, so $\approx 15\ m^2$

EXAMPLE 3

A car travels 120 km (to the nearest km) in 2.0 hours (to 1 d.p.). Find the greatest possible average speed.

1. $119.5 \le d < 120.5$

   Distance bounds.

2. $1.95 \le t < 2.05$

   Time bounds.

3. For the greatest speed: most distance, least time.

   A quotient — opposite bounds.

4. $v < \dfrac{120.5}{1.95} \approx 61.8\ km/h$

ANSWER

$\approx 61.8\ km/h$

03 Interactive model

Ask for the two extreme cases before any arithmetic — the rule then writes itself.

Which bound goes with which?

$a = 7 \pm 0.5$, $b = 4 \pm 0.5$. Largest $a + b$?

1. A sum grows when either part grows.

So both go up.

2. $UB(a) = 7.5$

, $UB(b) = 4.5$

3. $a + b < 12$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Upper bound	Yuqori chegara	Верхняя граница
Lower bound	Quyi chegara	Нижняя граница
Rounded to	Yaxlitlangan	Округлено до
Nearest	Eng yaqin	Ближайший
Half a unit	Birlikning yarmi	Половина единицы
Sum	Yig‘indi	Сумма
Difference	Ayirma	Разность
Product	Ko‘paytma	Произведение
Quotient	Bo‘linma	Частное

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$7\ cm$ to the nearest cm has bounds:

- 6.9 and 7.1
- 6.5 and 7.5
- 6 and 8
- 6.95 and 7.05

For the largest value of $a - b$ you take:

- both upper bounds
- both lower bounds
- UB(a) and LB(b)
- LB(a) and UB(b)

For the largest value of $a \div b$ you take:

- UB(a) ÷ UB(b)
- UB(a) ÷ LB(b)
- LB(a) ÷ LB(b)
- LB(a) ÷ UB(b)

$250\ g$ to the nearest 10 g has lower bound:

- 249.5
- 245
- 240
- 249

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *9 cm to the nearest cm. Give the bounds.*

   ANSWER $8.5 \le \ell < 9.5$

2. *3.2 m to 1 d.p. Give the bounds.*

   ANSWER $3.15 \le \ell < 3.25$

3. *80 kg to the nearest 10 kg. Give the bounds.*

   ANSWER $75 \le m < 85$

4. *2.50 s to 2 d.p. Give the bounds.*

   ANSWER $2.495 \le t < 2.505$

5. $a = 5 \pm 0.5$, $b = 3 \pm 0.5$. *Largest* $a + b$.

   ANSWER 9

6. *Same values: smallest* $a + b$.

   ANSWER 7

7. *600 m to the nearest 100 m. Give the bounds.*

   ANSWER $550 \le d < 650$

Medium · the standard question

7 PROBLEMS

1. $a = 7 \pm 0.5$, $b = 4 \pm 0.5$. Largest $a - b$.

   ANSWER 4

2. Same values: smallest $a - b$.

   ANSWER 2

3. A rectangle 4.2 m by 3.5 m (1 d.p.). Bounds of the perimeter.

   ANSWER $15.2 \le P < 15.6$

4. The same rectangle: bounds of the area.

   ANSWER $14.3175 \le S < 15.0875$

5. $d = 120$ km ± 0.5, $t = 2.0$ h ± 0.05. Largest speed.

   ANSWER ≈ 61.8 km/h

6. The same values: smallest speed.

   ANSWER ≈ 58.3 km/h

7. A square has side 8 cm to the nearest cm. Bounds of the area.

   ANSWER $56.25 \le S < 72.25$

Hard · stretch and reason

7 PROBLEMS

1. *A cube has edge 5 cm to the nearest cm. Bounds of its volume.*

   ANSWER $91.125 \le V < 166.375$

2. *To how many significant figures may that volume be given?*

   ANSWER None safely — the bounds do not even share the first digit; report "between 91 and 167 cm³".

3. $a = 12.4 \pm 0.05$, $b = 3.1 \pm 0.05$. *Bounds of* $\frac{a}{b}$

   ANSWER $\frac{12.35}{3.15} \approx 3.92$ to $\frac{12.45}{3.05} \approx 4.08$

4. *A triangle has base 9 cm and height 6 cm, both to the nearest cm. Bounds of its area.*

   ANSWER $\frac{1}{2} \cdot 8.5 \cdot 5.5 = 23.375$ to $\frac{1}{2} \cdot 9.5 \cdot 6.5 = 30.875$

5. *A runner covers 400 m (nearest m) in 52 s (nearest s). Bounds of the average speed.*

   ANSWER $\frac{399.5}{52.5} \approx 7.61$ to $\frac{400.5}{51.5} \approx 7.78$ m/s

6. *Explain why the upper bound uses < and the lower bound uses ≤.*

   ANSWER A value exactly on the lower bound rounds up to the stated figure; one exactly on the upper bound rounds up to the next figure instead.

7. *Two lengths are 5.0 cm and 5.00 cm. Which is stated more precisely, and why?*

   ANSWER *5.00 cm* — it claims the nearest *0.01*, so bounds *4.995* to *5.005*, ten times tighter.

07 Homework

Homework — 6 problems

Algebra 8, pp. 118–121, and Cambridge Learner's Book 3.4.

1. *15 cm to the nearest cm. Give the bounds.*
2. *6.70 kg to 2 d.p. Give the bounds.*
3. $a = 20 \pm 0.5$, $b = 8 \pm 0.5$. *Find the bounds of* $a - b$.

4. *A rectangle is 7.5 cm by 4.2 cm, both to 1 d.p. Find the bounds of the area.*
5. *A car covers 250 km (nearest km) in 4.0 h (1 d.p.). Find the bounds of the average speed.*
6. *A square has side 12 cm to the nearest cm. Find the bounds of the perimeter.*